

E014

Problem Solving for Technology Business Problems

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

After attending a CRM vendor's webinar, the VP of Sales at a company with offices in four cities says, "Our problem is that we need a new CRM." Which is the best problem statement for the team to work from?

- A. All four offices need the CRM from the webinar.
- B. Sales reps spend an average of 45 minutes a day re-entering customer data, delaying quotes by one day.
- C. The sales team is unhappy.
- D. We need a new CRM.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which statement is a solution disguised as a problem statement?

- A. "Order errors rose from 2% to 6% in the third quarter."
- B. "Password resets make up 30% of help-desk volume."
- C. "The problem is that we do not have a chatbot."
- D. "Customer wait times average 12 minutes against a 3-minute target."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A retailer's website, redesigned last year, slows sharply every Monday at 9 a.m. The team keeps adding web servers. Investigation shows a weekly reporting job runs against the production database at that exact time. The hosting contract renews in March. Which action addresses the root cause?

- A. Keep adding web servers.
- B. Reschedule or redirect the reporting job so it does not hit production at peak time.
- C. Renegotiate the hosting contract in March.
- D. Restart the servers every Monday at 8:55 a.m.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An 11-person team that deploys twice a week keeps having failed deployments. It has brainstormed dozens of possible causes and wants to sort them into categories such as people, process, tools, and environment. Which tool fits?

- A. A fishbone (Ishikawa) diagram.
- B. A Pareto chart.
- C. A control chart.
- D. A Gantt chart of the twice-weekly deployments.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

The help desk moved to a new ticketing tool in June. Last quarter's 900 tickets break down as: password resets 420, printer issues 180, VPN 150, email 90, other 60. The manager wants to address at least 60% of tickets by focusing on as few categories as possible. Which should she target?

- A. Password resets alone.
- B. Email and other issues.
- C. Password resets and printer issues (about 67%).
- D. All five equally, since every category matters the same.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Of 900 help-desk tickets, 420 are password resets and 180 are printer issues. What percentage of all tickets are password resets?

- A. 42%
- B. 70%
- C. About 214%
- D. About 46.7%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

HR and IT jointly manage employee onboarding. A DMAIC team has defined the problem and measured the baseline: new hires wait an average of 9 days for working accounts. The sponsor wants results by fall. What should the team do next?

- A. Analyze the process to find the root causes of the delay.
- B. Set up a control plan to monitor the process.
- C. Buy new onboarding software.
- D. Skip to Improve because the sponsor wants results by fall.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

A team's fix cut onboarding time from 9 days to 4. What should it do next under DMAIC?

- A. Control — set up monitoring and ownership so the gain holds.
- B. Declare the project finished.
- C. Analyze root causes again.
- D. Measure a new baseline from scratch.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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9.

During a holiday weekend, customers suddenly cannot complete payments on a company's website. The on-call engineer is new, and the team is still investigating the cause. What should the incident lead do about stakeholder communication?

- A. Wait until the root cause is confirmed before saying anything.
- B. Guarantee a complete fix within ten minutes.
- C. Send an initial update now with the impact, current actions, and when the next update will come.
- D. Update only IT leadership.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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10.

Departments that adopted a new collaboration tool, whose interface won a design award, show 15% higher productivity than those that did not. The same departments also got new managers and larger budgets this year. What can the team conclude?

- A. Roll the tool out company-wide immediately.
- B. Productivity rose because the interface won an award.
- C. The tool caused the 15% gain.
- D. The tool may be related to the gain, but other factors could explain it; a controlled comparison is needed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A team wants to test whether a collaboration tool actually causes higher productivity. Which approach gives the strongest evidence?

- A. Compare this year's adopters with last year's non-adopters.
- B. Give the tool to some comparable teams, chosen at random, and compare them with similar teams without it.
- C. Survey users on whether they like the tool.
- D. Rely on the correlation, since it always proves cause with a large sample.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A bank's innovation lab has a \$250,000 budget to improve its mobile app, built in 2019, for customers over 70. What should the team do first?

- A. Hold a workshop to brainstorm new features.
- B. Build and ship a prototype right away.
- C. Spend the budget on the latest AI features.
- D. Observe and interview older customers using the app.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

~~After attending a CRM vendor's webinar,~~ the VP of Sales ~~at a company with offices in four cities~~ says, "Our problem is that we need a new CRM." Which is the best problem statement for the team to work from?

Target word: well-formed problem statement **Decides:** It must describe a measurable gap, not a solution.

A	Camouflage	Uses the webinar and office count, which do not define the problem.
B	Correct	States an observable, measurable gap without prescribing a fix.
C	Dropped element	Drops any measure or specific gap.
D	Inverted operation	A solution stated as if it were the problem.

2. Answer C Concept 1 • transfer

Which statement is a solution disguised as a problem statement?

Target word: spot the disguised solution — reverse direction **Decides:** It names a tool rather than a gap.

A	Confusion set	A measured change in performance, not a solution.
B	Adjacent category	A factual observation that frames a problem.
C	Correct	It assumes the answer instead of describing a gap.
D	Inverted operation	A real, measurable gap.

3. Answer B Concept 2

~~A retailer's website, redesigned last year,~~ slows sharply every Monday at 9 a.m. The team keeps adding web servers. Investigation shows a weekly reporting job runs against the production database at that exact time. ~~The hosting contract renews in March.~~ Which action addresses the root cause?

Target word: root-cause fix **Decides:** The reporting job overloads the database.

A	Confusion set	Treats the symptom; the bottleneck is the database.
B	Correct	Removes the cause of the slowdown.
C	Camouflage	The contract date has nothing to do with the cause.
D	Adjacent category	A workaround that leaves the cause in place.

4. Answer A Concept 3

~~An 11-person team that deploys twice a week~~ keeps having failed deployments. It has brainstormed dozens of possible causes and wants to sort them into categories such as people, process, tools, and environment. Which tool fits?

Target word: tool for categorizing causes **Decides:** Organizing causes into categories.

A	Correct	It groups possible causes of one effect by category.
B	Adjacent category	Ranks causes by frequency; it needs counted data, not categories of ideas.
C	Confusion set	Tracks a process measure over time for variation.
D	Camouflage	Schedules tasks; the deployment frequency does not decide the tool.

5. Answer C Concept 4

~~The help desk moved to a new ticketing tool in June.~~ Last quarter's 900 tickets break down as: password resets 420, printer issues 180, VPN 150, email 90, other 60. The manager wants to address at least 60% of tickets by focusing on as few categories as possible. Which should she target?

Target word: vital few **Decides:** Password + printer = 600 of 900 $\approx 67\%$.

A	Dropped element	$420 / 900 \approx 47\%$, short of 60%.
B	Inverted operation	Picks the smallest categories.
C	Correct	The two largest categories pass 60% with the fewest picks.
D	Absolute language	Pareto analysis exists to focus on the vital few.

6. Answer D Concept 4 • transfer

Of 900 help-desk tickets, 420 are password resets and 180 are printer issues. What percentage of all tickets are password resets?

Target word: a single category's share **Decides:** $420 / 900$.

A	Wrong stem number	Treats the raw count 420 as if it were out of 1,000.
B	Confusion set	Divides by the top two (600) instead of all tickets.
C	Inverted operation	Divides 900 by 420.
D	Correct	$420 / 900 \approx 0.467$.

7. Answer A Concept 5

~~HR and IT jointly manage employee onboarding.~~ A DMAIC team has defined the problem and measured the baseline: new hires wait an average of 9 days for working accounts. ~~The sponsor wants results by fall.~~ What should the team do next?

Target word: next DMAIC phase **Decides:** Define and Measure are done.

A	Correct	Analyze follows Measure.
B	Inverted operation	Control is the last phase.
C	Dropped element	Skips analysis and jumps to a solution.
D	Camouflage	A deadline does not remove the need to find causes.

8. Answer A Concept 5 • transfer

A team's fix cut onboarding time from 9 days to 4. What should it do next under DMAIC?

Target word: phase after Improve **Decides:** Gains must be sustained.

A	Correct	Control sustains the improvement.
B	Dropped element	Drops the Control phase, so gains may slip.
C	Adjacent category	Analysis came before the fix.
D	Inverted operation	Measure is an earlier phase.

9. Answer C Concept 6

~~During a holiday weekend, customers suddenly cannot complete payments on a company's website. The on-call engineer is new, and the team is still investigating the cause. What should the incident lead do about stakeholder communication?~~

Target word: communication timing and content **Decides:** Impact is known; cause is not.

A	Inverted operation	Silence during a live outage erodes trust.
B	Absolute language	Promising an unknown fix time risks credibility.
C	Correct	Early, regular updates build trust even before the cause is found.
D	Dropped element	Drops the business stakeholders who are affected.

10. Answer D Concept 7

~~Departments that adopted a new collaboration tool, whose interface won a design award, show 15% higher productivity than those that did not. The same departments also got new managers and larger budgets this year. What can the team conclude?~~

Target word: valid conclusion **Decides:** Other changes happened at the same time.

A	Dropped element	Drops the confounders from the reasoning.
B	Camouflage	The award has no bearing on the data.
C	Confusion set	Treats correlation as causation.
D	Correct	Correlation with confounders does not show cause.

11. Answer B Concept 7 · transfer

A team wants to test whether a collaboration tool actually causes higher productivity. Which approach gives the strongest evidence?

Target word: design a causal test **Decides:** Only the tool should differ between groups.

A	Dropped element	Drops control over everything else that changed.
B	Correct	Random, comparable groups isolate the tool's effect.
C	Adjacent category	Measures satisfaction, not productivity.
D	Absolute language	Sample size does not remove confounders.

12. Answer D Concept 8

~~A bank's innovation lab has a \$250,000 budget to improve its mobile app, built in 2019, for customers over 70. What should the team do first?~~

Target word: first design-thinking step **Decides:** The team does not yet understand these users' needs.

A	Dropped element	Skips understanding the users.
B	Inverted operation	Prototyping comes after defining the need.
C	Camouflage	Budget size does not identify what users need.
D	Correct	Empathy comes first so the problem is defined from the user's view.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Problem statements describe a gap, not a solution	A good problem statement names the observable gap between current and desired performance, with a measure.	Stating a solution (“we need a new CRM”) as the problem.	A statement that names a product or tool.
2	Symptom vs. root cause	Fix the cause that produces the symptom; workarounds and added capacity only mask it.	Throwing resources at the symptom.	A recurring problem with a known trigger.
3	Fishbone (Ishikawa) diagram	A fishbone organizes possible causes of one effect into categories such as people, process, tools, and environment.	Confusing it with a Pareto chart, which ranks causes by frequency.	“Organize brainstormed causes into categories.”
4	Pareto analysis	Rank categories by frequency and focus on the vital few that make up most of the total.	Spreading effort evenly or starting with the easiest small categories.	Counts by category and a target share to address.
5	DMAIC sequence	Define, Measure, Analyze, Improve, Control — analyze causes before improving, and control to sustain gains.	Jumping from measurement straight to a solution, or stopping after the fix.	“The team has done X; what comes next?”
6	Incident communication	Communicate early with what is known, what is being done, and when the next update will come — even before the cause is found.	Waiting for the root cause or over-promising a fix time.	A live outage with an unknown cause.
7	Correlation is not causation	A relationship in data does not prove cause; rule out other factors with a controlled comparison.	Crediting the change you care about for an outcome with several possible causes.	Two things rising together, plus other changes at the same time.
8	Design thinking starts with users	Empathize and define the user’s need before generating ideas or prototypes.	Starting with a feature brainstorm or a technology.	An innovation effort aimed at a specific user group.

Course scope (public description): Problem Solving for Technology Business Problems invites students to use frameworks for resolving a range of issues that can emerge within an organization. Problem-solving frameworks offer a reliable, step-by-step process for pinpointing causes of difficulties and then methodically developing solutions. This course addresses business problems such as technology inefficiencies, process improvement and future innovation challenges, and incidents requiring immediate resolution and communication. There are no prerequisites for this course.